Stormwater Regulation Summary: Chicago, IL

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Combined sewer overflow reduction and flood control — Chicago has one of the largest combined sewer systems in the US.
- Chicago Stormwater Management Ordinance (2024 update) administered by DWM (Dept. of Water Management).
- Applies to ≥7,500 SF new/redeveloped impervious (regulated development threshold); O&M; plan required at ≥15,000 SF.
- Volume control storage for the first inch of runoff from impervious area, plus peak flow control for up to 100-yr storms.
- Green Permit Program fast-tracks permitting with fee waivers (up to $25,000) for buildings that incorporate green infrastructure.

2. Typical Soil Conditions

- Glacial lake plain soils — predominantly heavy clays (Hydrologic Soil Group D).
- Typical soils: Ashkum silty clay loam, Milford silty clay — very low infiltration rates (< 0.05 in/hr).
- Hydrologic Soil Group: Predominantly D across most of the city.
- Implication: Ground infiltration is extremely limited — Chicago regulations reflect this by emphasizing detention and controlled release rather than infiltration.
- Deep tunnel (TARP) system provides regional storage, but site-level management is still required.

3. Green Roof Requirements

- Green roofs are not legally mandatory but are strongly incentivized through the Green Permit Program.
- Chicago was an early US adopter — City Hall green roof installed 2001, catalyzed the market.
- Green Permit provides expedited review and fee waivers (up to $25,000) for projects that include green roofs and achieve LEED certification.
- Green roofs count toward the first-inch volume control requirement and receive stormwater fee credits.
- DWM recognizes green roofs as an approved BMP with quantified retention and detention credit.
- *The 2024 DWM regulation update may have adjusted incentive details — confirm current program terms.*

4. TSS Requirements

- *Chicago requires TSS removal for post-construction stormwater — verify specific percentage in 2024 DWM ordinance manual.*
- Green roofs are credited with TSS removal when properly designed — capturing the first inch of runoff addresses the highest-concentration pollutant load.
- First-flush capture inherently addresses a significant portion of the TSS load (first flush carries the highest pollutant concentrations).
- For runoff volumes exceeding green roof capacity, supplemental TSS treatment may be needed.
- DWM Stormwater Management Manual includes green roofs in approved BMP list with TSS credits.

5. Nature-Based Retention Requirements

- Volume control storage: retain the first inch of runoff from all impervious surfaces using approved BMPs.
- Approved retention BMPs include green roofs, bioretention, permeable pavement, and cisterns.
- CN (Curve Number) method is used for peak flow calculations; TR-55 methodology is standard.
- Typical urban CN: 98 (impervious); green roof reduces effective CN to approximately 65-75 depending on depth.
- Due to HSG D soils, volume-based retention credit (not infiltration credit) is the primary design approach.
- *2024 ordinance update may have refined retention calculation methodology — verify against current DWM guidance.*

6. Allowable Outflow Rates

- 2-year storm: Post-development peak flow ≤ pre-development peak flow.
- 100-year storm: Post-development peak flow ≤ pre-development peak flow.
- Chicago does not use a flat l/s/ha rate — control is relative to pre-development conditions.
- For reference, a typical Chicagoland undeveloped site (CN 70) produces approximately 20-35 l/s/ha for the 10-year storm.
- *Site-specific hydrology required; DWM reviews each project individually based on downstream sewer capacity.*